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DUAL UNLOCKING PADLOCK WITH DUAL INDICATORS

Abstract

The present invention is directed to a padlock including a body configured to house a combination mechanism and a key mechanism, the combination mechanism configured to allow operation of the padlock from a locked mode to an unlocked mode, and the key mechanism configured to allow operation of the padlock from the locked mode to an unlocked mode, a shackle having a short-leg and a long-leg, a key mechanism indicator configured to at least partially extend out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism, and a latch configured for operative engagement with the key mechanism indicator to retain the key mechanism indicator at least partially extended out of the body of the padlock until the padlock has been operated from the locked mode to the unlocked mode by the combination mechanism.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Appl. No. 63/563,216 filed Mar. 8, 2024, which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION
1. Field of the Invention
[0002] The present invention generally relates to padlocks, and more particularly to a dual unlocking padlock with a reset indicator which is enclosed in a locking body/housing.
2. Related Art
[0003] The combination padlock discussed in U.S. Pat. No. 7,140,209 entitled Padlock with Fully Integrated Dual Locking System, which is hereby incorporated by reference in its entirety, has a dual unlocking mechanism however, the padlock lacks an indication of when the shackle is in an open mode, a closed mode, or a reset mode. Therefore, the new invention seeks to address this issue by providing a padlock having a reset indicator such that the indicator provides a visual indication that the padlock is in the reset mode.
SUMMARY OF THE INVENTION
[0004] The new padlock also incorporates a key mechanism to override the combination mechanism like many TSA luggage locks which can let a TSA agent open the lock using the overriding key mechanism. The key mechanism user can open the lock via the key mechanism and a key mechanism indicator will pop upward indicating that the padlock has been opened via the key mechanism. The user can use the combination mechanism to reset the key mechanism indicator by entering the correct combination code of the padlock.
[0005] It is an object of the present invention to provide a padlock having a key-mechanism indicator provided therein that indicates the padlock has been opened via a key mechanism, and such key-mechanism indicator provides a visual indication that the padlock has been opened via the key mechanism until the correct combination code for the padlock is entered into a combination mechanism of the padlock such that the key-mechanism indicator cannot be depressed without entering the correct combination code.
[0006] It is another object of the present invention to provide a padlock having a recode indicator providing a visual indication as to when the padlock is in a reset mode, the reset mode allowing for resetting/reconfiguring the correct combination code for the combination mechanism of the padlock.
[0007] It is still another object of the present invention to provide a padlock so that only one of the key-mechanism indicator or the recode indicator are extended from the body of the padlock at one time.
[0008] It is yet another object of the present invention to provide a padlock where activation of the recode indicator may be preceded by resetting of the key-mechanism indicator.
[0009] It is another object of the present invention to provide a padlock having a recode/reset protrusion extending from a shackle that is configured for resetting of the key-mechanism indicator and activation of the recode indicator.
[0010] It is still another object of the present invention to provide a padlock in which the reset/recode protrusion could be separated into more than one protrusion where the reset protrusion is one protrusion and the recode protrusion is another protrusion.
[0011] It is yet another object of the present invention to provide a padlock configured for operation to a reset mode in which the correct combination code for the combination mechanism may be reset and/or reconfigured to a new correct combination code for the combination mechanism, and when the padlock is in the reset mode the user may have both hands free to rotate

the dials of the combination mechanism to the new correct combination code.

[0012] It is understood that these and other objects may be obtained in accordance with exemplary aspects of the present invention directed to a padlock that may include a body configured to house a combination mechanism and a key mechanism, the combination mechanism configured to allow operation of the padlock from a locked mode to an unlocked mode by the combination mechanism, and the key mechanism configured to allow operation of the padlock from the locked mode to an unlocked mode by the key mechanism.

[0013] According to this and other exemplary aspects of the present invention, the padlock may also include a shackle having a short-leg and a long-leg configured for retention in the body and for rectilinear and rotational movement relative to the body of the padlock.

[0014] According to this and other exemplary aspects of the present invention, the padlock may also include a key mechanism indicator configured to at least partially extend out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism.

[0015] According to this and other exemplary aspects of the present invention, the padlock may also include a latch configured for operative engagement with the key mechanism indicator to retain the key mechanism indicator at least partially extended out of the body of the padlock until the padlock has been operated from the locked mode to the unlocked mode by the combination mechanism.

[0016] According to this and other exemplary aspects of the present invention, the combination mechanism may include at least one dial having a plurality of indicia thereon, and at least one clutch configured for operative engagement with the at least one dial such that rotation of each of the at least one dial is transferred to the corresponding at least one clutch, each dial of the at least one dial may include a plurality of teeth, and each clutch of the at least one clutch may include an extended fin configured for operative engagement with the plurality of teeth of the corresponding dial from the at least one dial.

[0017] According to this and other exemplary aspects of the present invention, the key mechanism may include a cylinder configured for receipt of a correct cut key and a cam configured for operative engagement with the cylinder such that rotation of the cylinder is transferred to the cam, the cam may include a cam slope formed therein, and the padlock further may include a blocking plate having a pin extending therefrom and configured for operative engagement with the cam slope of the cam such that rotational movement of the cam is translated to rectilinear movement of the blocking plate.

[0018] According to this and other exemplary aspects of the present invention, the cam further may include a slope-wing extending therefrom, and the key-mechanism indicator may include a spring-loaded area formed on one end thereof, a bottom slope formed on an end of the key-mechanism indicator opposite the spring-loaded area, and a neck formed in the key-mechanism indicator disposed between the spring-loaded area and the bottom slope.

[0019] According to this and other exemplary aspects of the present invention, the padlock further may include a key-mechanism indicator spring configured for operative engagement with the spring-loaded area of the key-mechanism indicator to urge the key-mechanism indicator into the body of the padlock, and the slope-wing is configured for operative engagement with the bottom slope of the key-mechanism indicator such that rotation of the cam causes compression of the key-mechanism indicator spring and extension of at least a portion of the key-mechanism indicator out of the body of the padlock.

[0020] According to this and other exemplary aspects of the present invention, the latch may include a spring assembled area configured for operative engagement with a latch spring and a tip extending from the spring assembled area, the tip is configured for a retaining engagement with the neck of the key-mechanism indicator so as to maintain extension of at least the portion of the key-mechanism indicator out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism.

[0021] According to this and other exemplary aspects of the present invention, the padlock may also include a locking hole formed in the body and configured to at least partially surround the short-leg of the shackle when the padlock is in the locked mode, the locking hole has an arcuate opening formed therein configured for receipt of at least a portion of the blocking plate when the padlock is in the locked mode, the blocking plate is rectilinearly positionable between an extended position when the padlock is in the locked mode and a retracted position when the padlock is in the unlocked mode by the key mechanism as a result of rotational movement of the cam, and when the blocking plate is in the retracted position the shackle may be rotated about the long-leg of the shackle to remove the short-leg of the shackle from the locking hole and operate the padlock to the unlocked mode by the key mechanism.
[0022] According to this and other exemplary aspects of the present invention, the cylinder may include a square slot, and the cam further may include a square block configured to be received within the square slot of the cylinder so that rotational movement of the cylinder is transferred to the cam.
[0023] According to this and other exemplary aspects of the present invention, the blocking plate further may include a blocking-edge, when the blocking plate is in the extended position the blocking-edge is positioned in the arcuate opening of the locking hole to prevent movement of the short-leg of the shackle through the arcuate opening of the locking hole, and when the blocking plate is in the retracted position the blocking-edge is removed from the arcuate opening of the locking hole.
[0024] According to this and other exemplary aspects of the present invention, the long-leg of the shackle may include a protrusion extending therefrom for each clutch of the at least one clutches, and each clutch of the at least one clutches further may include a shackle hole configured to receive the long-leg of the shackle through the clutch, a top surface positioned within the shackle hole, a bottom surface positioned opposite the top surface and an opening gap extending from the top surface to the bottom surface and forming a passageway therebetween for passage of the protrusion of the long-leg of the shackle corresponding to the clutch at least partially through the clutch.
[0025] According to this and other exemplary aspects of the present invention, the padlock may also include a locking hole formed in the body and configured to at least partially surround the short-leg of the shackle when the padlock is in the locked mode, rectilinear movement of the long-leg of the shackle away from the body of the padlock is configured to remove the short-leg of the shackle from the locking hole when the padlock is operated to the unlocked mode by the combination mechanism, and the padlock is in the unlocked mode by the combination mechanism when a correct combination code has been set on all of the dials of the at least one dial and the opening gap of each clutch are aligned with each corresponding protrusion of the long-leg of the shackle allowing for movement of each protrusion from the corresponding bottom surface to the corresponding top surface of each corresponding clutch.
[0026] According to this and other exemplary aspects of the present invention, the long-leg of the shackle further may include a recode protrusion extending therefrom, when the padlock is in the unlocked mode by the combination mechanism the recode protrusion is configured for alignment with a notch formed in the body and insertion of the recode protrusion into the notch engages each protrusion of the long-leg of the shackle with the corresponding top surface of each corresponding clutch thereby causing separation of the extended fin of each clutch from the teeth of the corresponding dial, and the extended fin of each clutch is retained in a fin slot within the body.
[0027] According to this and other exemplary aspects of the present invention, the padlock may also include a recode indicator having a shackle receiving hole configured for receipt of the long-leg of the shackle, a spring loaded area configured for insertion in a shackle spring and a clutch contact surface, insertion of the recode protrusion into the notch and engagement of each protrusion of the long-leg of the shackle with the corresponding top surface of each corresponding clutch causes operative contact between the bottom surface of at least one clutch with the clutch contact

surface and depression of the shackle spring, and depression of the shackle spring allows for at least partial extension of the spring loaded area of the recode indicator out of the body of the padlock.

[0028] According to this and other exemplary aspects of the present invention, the long-leg of the shackle further may include a reset protrusion extending therefrom, the key-mechanism indicator may include a spring-loaded area formed on one end thereof and a neck formed in the key-mechanism indicator, the padlock further may include a key-mechanism indicator spring configured for operative engagement with the spring-loaded area of the key-mechanism indicator to urge the key-mechanism indicator into the body of the padlock, the latch may include a spring assembled area configured for operative engagement with a latch spring, a tip extending from the spring assembled area and a tail slope extending from the latch opposite the tip, the tip is configured for a retaining engagement with the neck of the key-mechanism indicator so as to maintain extension of at least the portion of the key-mechanism indicator out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism, and the reset protrusion is configured for operative engagement with the tail slope of the latch to urge the latch to compress the latch spring and disengage the tip from the neck of the key-mechanism indicator when the padlock is operated to the unlocked mode by the combination mechanism.

[0029] According to this and other exemplary aspects of the present invention, at least partial extension of the spring loaded area of the recode indicator out of the body of the padlock provides a visual indication the padlock has been operated to a reset mode, the in reset mode rotation of each dial is not transferred to the clutch corresponding to each of the at least one dials.

[0030] According to this and other exemplary aspects of the present invention, the body may include at least one recode channel formed therein integrally connected with the notch, the recode channel is configured for receipt and retention of the recode protrusion of the long-leg of the shackle within the body after the recode protrusion has been inserted through the notch, and when the recode protrusion is retained within the body by the recode channel separation of the extended fin of each clutch from the teeth of the corresponding dial is maintained.

[0031] According to this and other exemplary aspects of the present invention, the long-leg of the shackle further may include a reset protrusion extending therefrom, the key-mechanism indicator may include a spring-loaded area formed on one end thereof and a neck formed in the key-mechanism indicator, the padlock further may include a key-mechanism indicator spring configured for operative engagement with the spring-loaded area of the key-mechanism indicator to urge the key-mechanism indicator into the body of the padlock, the latch may include a spring assembled area configured for operative engagement with a latch spring, a tip extending from the spring assembled area and a tail slope extending from the latch opposite the tip, the tip is configured for a retaining engagement with the neck of the key-mechanism indicator so as to maintain extension of at least the portion of the key-mechanism indicator out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism, and the reset protrusion is configured for operative engagement with the tail slope of the latch to urge the latch to compress the latch spring and disengage the tip from the neck of the key-mechanism indicator when the padlock is operated to the unlocked mode by the combination mechanism.

[0032] According to this and other exemplary aspects of the present invention, the long-leg of the shackle further may include a recode protrusion extending therefrom, when the padlock is in the unlocked mode by the combination mechanism the recode protrusion is configured for alignment with a notch formed in the body and insertion of the recode protrusion into the notch engages each protrusion of the long-leg of the shackle with the corresponding top surface of each corresponding clutch thereby causing separation of the extended fin of each clutch from the teeth of the corresponding dial, and the extended fin of each clutch is retained in a fin slot within the body.

[0033] According to this and other exemplary aspects of the present invention, the key-mechanism indicator may include a spring-loaded area formed on one end thereof and a neck formed in the key-mechanism indicator, the padlock further may include a key-mechanism indicator spring configured for operative engagement with the spring-loaded area of the key-mechanism indicator to urge the key-mechanism indicator into the body of the padlock, the latch may include a spring assembled area configured for operative engagement with a latch spring, a tip extending from the spring assembled area and a tail slope extending from the latch opposite the tip, the tip is configured for a retaining engagement with the neck of the key-mechanism indicator so as to maintain extension of at least the portion of the key-mechanism indicator out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism, and the recode protrusion is configured for operative engagement with the tail slope of the latch to urge the latch to compress the latch spring and disengage the tip from the neck of the key-mechanism indicator when the padlock is operated to the unlocked mode by the combination mechanism.

[0034] According to this and other exemplary aspects of the present invention, when the padlock is in the unlocked mode by the combination mechanism the reset protrusion is configured for alignment with a notch formed in the body and insertion of the reset protrusion into the notch engages each protrusion of the long-leg of the shackle with the corresponding top surface of each corresponding clutch thereby causing separation of the extended fin of each clutch from the teeth of the corresponding dial, and the extended fin of each clutch is retained in a fin slot within the body.

[0035] According to this and other exemplary aspects of the present invention, the padlock may also include a locking hole formed in the body and configured to at least partially surround the short-leg of the shackle when the padlock is in the locked mode, rectilinear movement of the long-leg of the shackle away from the body of the padlock is configured to remove the short-leg of the shackle from the locking hole when the padlock is operated to the unlocked mode by the combination mechanism, and rotational movement of the shackle about the long-leg of the shackle is configured to remove the short-leg of the shackle from the locking hole when the padlock is operated to the unlocked mode by the key mechanism.

[0036] The reference numbers and corresponding elements are summarized below:

TABLE-US-00001 10 Padlock. 20 Front body 21 Locking hole 22 Long-leg shackle hole 23 Cylinder hole 24 Cam hole 25 Latch member cutout 26 Clutch hole 27 Dial hole 28 Recode indicator hole 29 Wall 20a Fin-slot 20b Recode channel 30 Rear body 31 Locking hole 32 Long-leg shackle hole 33 Cylinder hole 34 Cam hole 35 Key-mechanism indicator hole 36 Latch member cutout 37 Clutch hole 38 Dial hole 39 Recode indicator hole 30a Wall 30b Notch 30c Recode channel 30d Fin-slot 40 Cylinder 41 Square slot 50 Cam 51 Square block 52 Slope 53 Slope-wing 60 Blocking plate 61 Blocking-edge 62 Pin 70 Key-mechanism indicator 71 Spring loaded area 72 Neck 73 Bottom slope 80 Latch 81 Spring assembled area 82 Tip 83 Tail-slope 90 Clutch 91 Extended Fin 92 Opening-gap 93 Shackle hole 94 Top surface 95 Bottom surface 100 Dial 101 Teeth 102 Cutout 110 Recode indicator 111 Spring loaded area 112 Large diameter 113 Contact clutch surface 114 Shackle hole receiving hole 120 Shackle 121 Short-leg-shackle 122 Long-leg-shackle 123 Protrusion 124 Recode/Reset-Protrusion 130 Dial Spring Plate 140 Shackle Spring 150 Latch spring 160 Key-mechanism indicator spring. 170 Key

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Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0037] For a fuller understanding of the nature and object of the present invention, reference should be had to the following detailed description taken in connection with the accompanying drawings, in which:

[0038] FIG. **1**A is a front view of an exemplary padlock according to an aspect of the invention in a locked mode with an exemplary front body removed;

[0039] FIG. **1**B is a rear view of the exemplary padlock with an exemplary rear body removed;

[0040] FIG. **1**C is a perspective front view of the exemplary padlock with the exemplary front body removed;
[0041] FIG. **1**D is a perspective rear view of the exemplary padlock with the exemplary rear body removed;
[0042] FIG. **2** is a perspective view of the exemplary front body that may be used with the exemplary padlock according to an aspect of the present invention;
[0043] FIG. **3** is a perspective view of the exemplary rear body that may be used with the exemplary padlock according to an aspect of the present invention;
[0044] FIG. **4** is a perspective view of an exemplary cylinder that may be used with the exemplary padlock according to an aspect of the present invention;
[0045] FIG. **5** is a perspective view of an exemplary cam that may be used with the exemplary padlock according to an aspect of the present invention;
[0046] FIG. **6** is a perspective view of an exemplary blocking plate that may be used with the exemplary padlock according to an aspect of the present invention;
[0047] FIG. **7** is a perspective view of an exemplary key-mechanism indicator that may be used with the exemplary padlock according to an aspect of the present invention;
[0048] FIG. **8** is a perspective view of an exemplary latch that may be used with the exemplary padlock according to an aspect of the present invention;
[0049] FIG. **9** is a perspective view of an exemplary clutch that may be used with the exemplary padlock according to an aspect of the present invention;
[0050] FIG. **10** is a perspective view of an exemplary dial that may be used with the exemplary padlock according to an aspect of the present invention;
[0051] FIG. **11** is a perspective view of an exemplary recode indicator that may be used with the exemplary padlock according to an aspect of the present invention;
[0052] FIG. **12** is a perspective view of an exemplary shackle that may be used with the exemplary padlock according to an aspect of the present invention;
[0053] FIG. **13**A is a front view of the exemplary padlock according to an aspect of the invention in an unlocked mode by a key mechanism with the exemplary front body removed;
[0054] FIG. **13**B is a rear view of the exemplary padlock with an exemplary rear body removed;
[0055] FIG. **13**C is a perspective front view of the exemplary padlock with the exemplary front body removed;
[0056] FIG. **13**D is a perspective rear view of the exemplary padlock with the exemplary rear body removed;
[0057] FIG. **14**A is a front view of the exemplary padlock according to an aspect of the invention in an unlocked mode by a combination mechanism with the exemplary front body removed;
[0058] FIG. **14**B is a rear view of the exemplary padlock with an exemplary rear body removed;
[0059] FIG. **14**C is a perspective front view of the exemplary padlock with the exemplary front body removed;
[0060] FIG. **14**D is a perspective rear view of the exemplary padlock with the exemplary rear body removed;
[0061] FIG. **15**A is a front view of the exemplary padlock according to an aspect of the invention in a reset mode with the exemplary front body removed;
[0062] FIG. **15**B is a rear view of the exemplary padlock with an exemplary rear body removed;
[0063] FIG. **15**C is a perspective front view of the exemplary padlock with the exemplary front body removed;
[0064] FIG. **15**D is a perspective rear view of the exemplary padlock with the exemplary rear body removed;
[0065] FIG. **16**A is a front view of the exemplary padlock according to an aspect of the invention in a reset key mechanism indicator mode with the exemplary front body removed;
[0066] FIG. **16**B is a rear view of the exemplary padlock with an exemplary rear body removed;

[0067] FIG. **16**C is a perspective front view of the exemplary padlock with the exemplary front body removed; and

[0068] FIG. **16**D is a perspective rear view of the exemplary padlock with the exemplary rear body removed.

DETAILED DESCRIPTION

[0069] The present invention now will be described more fully hereinafter with reference to the accompanying figures, in which exemplary embodiments of the invention are shown. The invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Like reference numerals refer to like elements throughout.

[0070] Referring now to FIGS. **1**A-**16**D, therein illustrated is an exemplary padlock, generally indicated by reference numeral **10** in FIGS. **1**A-**1**D, **13**A-**13**D, **14**A-**14**D, **15**A-**15**D and **16**A-**16**D. The padlock **10** may be constructed from a front body **20** and a rear body **30** that may be combined together in order to form a housing for the components of the padlock **10**. The padlock **10** may also include a shackle **120** configured for retention within the front body **20** and the rear body **30**. The front body **20** of the padlock **10** may include a locking hole **21** positioned so as to be opposite a corresponding locking hole **31** of the rear body **30**, and form an opening configured to receive and releasably retain a short-leg shackle **121** of the shackle **120**. The front body **20** may also include a long-leg shackle hole **22** positioned so as to be opposite a long-leg shackle hole **32** of the rear body **30** and form an opening configured to receive and retain a long-leg shackle **122** of the shackle **120**. Each of the front body **20** and the rear body **30** include a cylinder hole **23**/**33** configured and dimensioned to receive and retain a cylinder **40** within the padlock **10**. The front body **20** and the rear body **30** may each further include a cam hole **24**/**34** configured and dimensioned to receive and retain a cam **50** operatively coupled to the cylinder **40** within the padlock **10**. The front body **20** and the rear body **30** may each further include a latch member cutout **25**/**36** configured and dimensioned to receive and retain a cam latch **80** within the padlock **10**. The rear body **30** may further include a key-mechanism indicator hole **35** configured and dimensioned to receive and retain a key-mechanism indicator **70** within the padlock **10**. The front body **20** and the rear body **30** may also each include one or more clutch holes **26**/**37**, one or more dial holes **27**/**38** and one or more fin-slots **20***a*/**30***d*. The one or more clutch holes **26**/**37** are configured and dimensioned to each receive one of the one or more clutches **90** within the padlock **10**, and the one or more dial holes **27**/**38** are configured and dimensioned to each receive one of the one or more dials **100** within the padlock **10**. The one or more clutches **90** each include at least one extended-fin **91**, and each extended-fin **91** configured to be received within a corresponding fin-slot **20***a*/**30***d* of the one or more fin-slots **20***a*/**30***d* when a correct combination code of the padlock **10** has been set on the one or more dials **100**.

[0071] Still referring to FIGS. **1**A-**16**D, each of the front body **20** and the rear body **30** of the padlock **10** may also each include a recode indicator hole **28**/**39** configured to allow a recode indicator **110** of the padlock **10** to extend at least partially out of the padlock **10**. The recode indicator **110** may include a substantially cylindrical spring loaded area **111**, a large diameter **112** extending circumferentially from the spring loaded area **111** and configured to separate a contact clutch surface **113** from the spring loaded area. The recode indicator **110** may also include a shackle hole receiving hole **114** configured to receive an end of the long-leg shackle **122**. The spring loaded area **111** is configured to be received within shackle spring **140**, and the shackle spring **140** is configured to engage with the larger diameter **112** to urge the recode indicator **110** in a direction into the padlock **10**. The front body **20** and the rear body **30** may also each include a wall **29**/**30***a* configured to restrict upward movement of the one or more clutches **90** within the padlock **10**. The wall **29**/**30***a* is configured and positioned so that an uppermost protrusion **123** of the one or more protrusions **123** of the shackle **120** contact the wall **29**/**30***a* when the padlock **10** has been operated to an unlocked mode via the combination mechanism in order to restrict further movement of the shackle **120** away from the front body **20** and the rear body **30**. Likewise, the wall

29/**30***a* may also be configured to restrict upward movement of the topmost clutch **90** of the one or more clutches **90**. The front body **20** and the rear body **30** may also each include a recode channel **20***b*/**30***c* configured to engage with a recode/reset protrusion **124** of the shackle **120**. The recode channel **30***c* formed in the rear body **30** allows insertion of the recode/reset protrusion **124** into the padlock **10** after the padlock **10** has been operated to the unlocked mode by the combination mechanism.

[0072] Still referring to FIGS. **1**A-**16**D, the cylinder **40** of the padlock **10** may include a square slot **41** configured to receive a square block **51** of the cam **50** so that when a correct cut key **170** is inserted into the cylinder **40** and rotated, rotational movement of the cylinder **40** is transferred to the cam **50**. The cam **50** may further include a slope **52** formed therein and extending at an angle relative to the square block **51** along the periphery of the cam **50**. The padlock **10** may further include a blocking plate **60** having a blocking edge **61** configured for positioning in the locking hole **31** of the rear body **30** in order to restrict and/or prevent rotational movement of the short-leg shackle **121** out of the locking hole **21**/**31**. The blocking plate **60** may further include a pin **62** extending therefrom, and configured for engagement with the slope **52** of the cam **50** so that rotational movement of the cam **50** is transferred to rectilinear movement of the blocking pate **60** out of the locking hole **31** when the correct cut key **170** has been inserted and rotated in the cylinder **40**. The cam **50** may also include a slope-wing **53** configured for engagement with a bottom slope **73** of the key-mechanism indicator **70**. As the cam **50** is rotated by rotational movement of the correct cut key **170** in the cylinder **40**, the slope-wing **53** engages with the bottom slope **73** of the key-mechanism indicator **70** to urge the key-mechanism indicator **70** against a key-mechanism spring **160** configured to urge the key-mechanism indicator **70** into a retracted position within the padlock **10**. The key-mechanism spring **160** engages with a spring loaded area **71** of the key-mechanism indicator **70**. As the cam **50** is rotated, the slope-wing **53** also rotates to urge the key-mechanism indicator **70** so that a neck **72** formed in the key-mechanism indicator **70** is engaged with a tip **82** of the latch **80**. The latch **80** may also include a spring assembled area **81** configured for engagement with a latch spring **150**. The latch spring **150** urges the tip **82** of the latch **80** into engagement with the neck **72** of the key-mechanism indicator **70** so that the key-mechanism spring **160** is compressed and the key-mechanism indicator **70** extends from the padlock **10**. The latch **80** may also include a tail-slope **83** which is positioned and configured for engagement with the recode/reset protrusion **124** of the shackle **120**. Each of the one or more dials **100** may have teeth **101** positioned along an interior perimeter of the dials **100**, which teeth **101** are configured for engagement with the one or more extended-fins **91** of the one or more clutches **90** so that rotation of the dials **100** around the long-leg shackle **122** results in corresponding rotation of the clutches **90**. The extended-fins **91** are assembled in the teeth **101** of the dials **100** so that turning of dials **100** will turn the clutches **90**. Each of the dials **100** has one or more indicia, which may be in the form of numbers, symbols, letters, colors or the like, corresponding to each tooth of the teeth **101**. For example, in the exemplary embodiment illustrated ten indicia in the form of the numbers 0-9 are illustrated, and each of the dials **100** has ten teeth **101**. A dial spring plate **130** may be positioned in the padlock **10** in order to give an indication when a distinct indicia of the combination controlled mechanism has been selected as the dial spring plate **130** will engage with cutouts **102** on the dials **100** positioned between each indicia. Each of the clutches **90** has a substantially cylindrical configuration having a bottom surface **95**. Within each of the clutches **90** is a shoulder forming a top surface **94**, an opening gap **92** formed in the top surface **94**, and a shackle hole **93** formed through the clutch **90**. The long-leg shackle **122** of the shackle **120** includes one or more protrusions **123** extending therefrom, and each configured for engagement with the bottom surface **95** of one of the one or more clutches **90** when the padlock **10** is in a locked mode, or configured to pass through the opening gap **92** in one of the one or more clutches **90** when the padlock **10** is in the unlocked mode by the combination mechanism.

[0073] Referring now to FIGS. **1**A-**12**, the locked mode of the padlock **10** will now be discussed.

The padlock **10** contains the shackle **120** having the short-leg-shackle **121** of the shackle **120** of the padlock **10** may be controlled between a closed mode and an open mode by the key mechanism, which may be comprised of the cylinder **40**, the came **50** and the blocking plate **60**. In the closed mode of the short-leg-shackle **121** the padlock **10** is in the locked mode, and then the short-leg-shackle **121** is in the open mode by the key mechanism, the padlock **10** may be operated to the unlocked mode by the key mechanism. The long-leg-shackle **122** of the shackle **120** may be controlled by the combination mechanism, which may be comprised of the one or more clutches **90** and the one or more dials **100**, to allow for operation of the padlock between the locked mode and the unlocked mode by the combination mechanism of the padlock **10**. When the padlock **10** is in the locked mode, the shackle **120** will have no rectilinear movement away from the padlock **10**, and the short-leg-shackle **121** is retained in the locking-hole **21**/**31** of the front/rear body **20**/**30**. The short-leg shackle **121** is retained in the locking-hole **21**/**31** by the blocking plate **60** preventing rotation of the short-leg shackle **121** away from the locking-hole **21**/**31**. As there is no correct cut key **170** inserted into the cylinder **40**, the cam **50** cannot rotate, and the blocking plate **60** keeps the short-leg-shackle **121** in the closed mode and prevents the padlock **10** from operating to the unlocked mode by the key mechanism. Furthermore, when the padlock **10** is in the locked mode, at least one dial **100** is not set to the correct combination code, which results in at least one of the corresponding clutches **90** not having the opening-gap **92** aligned with the corresponding protrusion **123** of the long-leg shackle **122**. Accordingly, the shackle **120** cannot be moved in a direction away from the padlock **10** in order to release the short-leg shackle **121** from the locking hole **21**/**31**. In the locked mode, at least one protrusion **123** of the shackle **120** is not aligned with the opening-gap **92** of the clutch **90**, and the shackle **120** cannot be lifted upward to operate the padlock **10** to the unlocked mode by the combination mechanism, because the bottom surface **95** of each clutch **90** corresponding to a dial **100** not containing the correct combination code restricts rectilinear movement of the long-leg-shackle **122**.

[0074] In the locked mode, the correct cut key **170** is not present and therefore, the cylinder **40** can have no rotational movement. The cylinder **40** may be a pin-tumbler cylinder, a wafer cylinder or any other type of cylinder known in the art, and it is understood that when the correct cut key **170** is not inserted into the cylinder **40**, the pins, wafers, or the like are not correctly aligned and inhibit rotational movement of the cylinder **40**. The pin **62** of the blocking plate **60** is assembled onto the slope **52** of the cam **50**. Since the cylinder **40** cannot rotate, the slope **52** of the cam **50** likewise has no rotational movement, and the pin **62** of the blocking plate **60** will be retained in place by the slope **52** of the cam **50**. It is understood that the blocking plate **60** cannot have any rectilinear movement unless the cam **50** has rotational movement, and the blocking plate **60** is in the locked position. As the cylinder **40** has no movement, the cam **50** will also have no rotational movement as a result of engagement between the square slot **41** of the cylinder **40** and the square block **51** of the cam **50**. When the blocking plate **60** in the locked position, the blocking edge **61** of the blocking plate **60** will block the path of the short-leg shackle **121** from moving out of the locking-hole **21**/**31** of the lock body **20**/**30**, and the padlock **10** is in the locked mode.

[0075] Referring now to FIGS. **13**A-**13**D, the operation of the padlock **10** to the unlocked mode by the key mechanism will now be discussed. Using the key mechanism to unlock the padlock **10**, the correct cut key **170** is inserted into the cylinder **40**, and rotated so that the cam **50** will rotate in the same manner as the cylinder **40** due to the operative coupling of the square slot **41** with the square block **51** of the cam **50**. As the cylinder rotates 40, the cam **50** rotates, and the slope-wing **53** of the cam **50** will contact the bottom slope **73** of the key-mechanism indicator **70**. The slope-wing **53** will push the bottom slope **73** of the key-mechanism indicator **70** upward, and the key-mechanism indicator **70** will also be pushed upward. As the key-mechanism indicator **70** travels upward, the neck **72** of the key-mechanism indicator **70** algins with the tip **82** of the latch **80**. The latch spring **150** will push the tip **82** of the latch **80** toward the neck **72** of the key-mechanism indicator **70** so that the tip **82** becomes engaged with the neck **72**. The key-mechanism indicator **70** will not be able to travel downward in such engaged position, and the key-mechanism indicator **70** will remain extended from the padlock **10**. In this manner, the padlock **10** provides a visual indication that the padlock **10** has been opened by the correct cut key **170**, for example, as a result of baggage inspection by an official inspector at an airport. To operate the padlock **10** to the unlocked mode by the key mechanism, the cam **50** is rotated as a result of rotation of the correct cut key **170** in the cylinder, and rotation of the cam **50** will be transferred to rectilinear movement of the blocking plate **60** due to the interaction of the slope **52** of the cam **50** with the pin **62** of the blocking plate **60**. The pin **62** of the blocking plate **60** is configured to travel along the slope **52** of the cam **50**, so as the cam **50** is rotated the pin **62** retains its location relative to the slope **52** and the position of the blocking plate **60** relative to the cam **50** is adjusted depending upon the location of the pin **62** along the slope **52** of the cam **50**. As the blocking plate **60** becomes retracted relative to the cam **50**, the blocking-edge **61** of the blocking-plate **60** moves away from the locking-hole **21**/**31** of the body **20**/**30**. The short-leg-shackle **121** of the padlock **10** can be operated to the open mode by rotating the short-leg-shackle **121** out of the locking-hole **21**/**31** of the body **20**/**30** and thereby operating the padlock **10** to the unlocked mode by the key mechanism. If the user wants to position the short-leg-shackle **121** of the shackle **120** back into the closed mode, and return the padlock **10** to the locked mode, the user can rotate the short-leg-shackle **121** of the shackle **120** to the locking-hole **21**/**31** of the body **20**/**30**, and use the correct cut key **170** to rotate the cylinder **40** to the locked position by counter rotating the cylinder **40**, and then the slope **52** of the cam **50** will drag the pin **62** of the blocking-plate **60** upward causing extension of the blocking plate **60** relative to the cam **50**. As the blocking plate **60** is operated to its original position relative to the cam **50**, the blocking-edge **61** of the blocking plate **60** will block the locking hole **21**/**31** of the body **20**/**30**. The user can withdraw the key **170**. Therefore, the short-leg **121** of the shackle **120** will be in the closed mode, and the padlock **10** will be in the locked mode.

[0076] Referring now to FIGS. **14**A-**14**D and **16**A-**16**D, operation of the padlock **10** to the unlocked mode by the combination mechanism, and resetting of the key-mechanism indicator **70** will now be discussed. When using the combination mechanism to operate the padlock **10** to the unlocked mode by the combination mechanism, the fins **91** of each of the clutches **90** are fully engaged with the teeth **101** of the corresponding dials **100**. The user will algin dials **100** to the correct combination code and the opening-gap **92** of each clutch **90** will algin to the corresponding protrusion **123** on the long-leg shackle **122** of the shackle **120**. Each of the clutches **90** contains a shackle hole **93** to receive the long-leg shackle **122**. The dial spring plate **130** engages with the cutout **102** of the dial **100** to make sure each dial **100** contains a ratchet feeling to make sure the dial **100** will have a feeling for each indicia on the dial **100**. When all of the protrusions **123** on the long-leg shackle **122** are aligned with the corresponding opening-gaps **92** there is nothing to block the protrusions **123** of the shackle **120**, and the shackle **120** may have rectilinear movement within the body **20**/**30** of the padlock **10**. The protrusions **123** are no longer blocked by the bottom surface **95** of the corresponding clutch **90**, and the shackle **120** can move in a direction away from the body **20**/**30** so that the short-leg **121** of the shackle **120** may be moved away from the locking-hole **21**/**31** of the lock body **20**/**30** to the open mode. The shackle **120** can be moved away from the lock body **20**/**30** until the uppermost protrusion **123** on the long-leg-shackle **122** contacts the wall **29**/**30***a* of the lock body **20**/**30**. Once the shackle **120** is operated to the open mode then the shackle **120** will be free to rotate about the long-leg shackle **122**. To position the padlock **10** into the locked mode, the user can push the shackle **120** back towards the padlock **10** such that the short-leg **121** will be reinserted into the locking-hole **21**/**31** of the lock body **20**/**30**. The blocking-edge **61** of the blocking plate **60** will prevent the short-leg **121** of the shackle **120** from rotating away from the locking hole **21**/**31**, because the key lock mechanism is in the locked position then the blocking plate **60** will remain in locked position. This will make the blocking-plate **60** restrict the shackle **120** from rotating to the open mode. As the shackle **120** is in the locked position then the protrusions **123** of the long-leg-shackle **122** will be positioned below the corresponding bottom surfaces **95** of each of the clutches **90**. The user can then scramble the dials **100** so that the dials **100** are no longer set to the correct combination code, and the fins **91** of each of the one or more clutches **90** will rotate in the same manner such that the opening-gaps **92** of the clutches **90** will not align with the corresponding protrusions **123** of the long-leg-shackle **122** of the shackle **120**. In such case, shackle **120** will no longer be able to be pulled away from the padlock **10**, and the padlock **10** is now returned to the locked mode.

[0077] Referring more specifically to FIGS. **16**A-**16**D, in order to reset the key-mechanism indicator **70**, the user operates the padlock **10** to the unlocked mode by the combination mechanism as discussed above. Then the user pulls the shackle **120** upward so that the recode/reset protrusion **124** of the shackle **120** will contact the tail-slope **83** of the latch **80**. The contact of the recode/reset-protrusion **124** and the tail-slope **83** will make the latch **80** move leftward (an opposition of the spring force of the latch spring **150**), and the tip **82** of the latch **80** will disengage from the neck **72** of the key-mechanism indicator **70**. Then the key-mechanism indicator spring **160** engaged with the spring load area **71** of the key-mechanism indicator **70** will push the indicator **70** down back to the original position retracted and hidden in the lock body **20**/**30** of the padlock **10**.

[0078] Referring now to FIGS. **15**A-**15**D, the reset mode of the padlock **10** will now be discussed in which the correct combination code for the combination mechanism can be changed and/or reset. In the reset mode, the user operates the padlock **10** to the unlocked mode by the combination mechanism as discussed above with respect to FIGS. **14**A-**14**D. As the shackle **120** is pulled upward, the user can turn the shackle **120** ninety (90) degrees such that the recode/reset-protrusion **124** will align to the notche **30***b* of the rear body **30** such that shackle **120** can be pushed downward so that each of the protrusions **123** of the shackle **120** will push the top surfaces **94** of each of the the clutches **90** downward. As the user pushes the shackle **120** completely downward, the recode/reset-protrusion **124** of the shackle **120** would be able to turn the recode/reset-protrusion **124** to the recode channel **20***b*/**30***c* of the lock body **20**/**30**. The recode channel **20***b*/**30***c* will only allow the recode/reset-protrusion **124** to engage and disallow the shackle **120** from having upward movement. The recode channel **20***b*/**30***c* allows the user to have both hands free without further requiring the user to keep pushing the shackle **120** downward for the entire reset process. As the set of clutches **90** will be pushed downward, the fins **91** on each of the clutches **90** will disengage from the teeth **101** of the corresponding dials **100**. The user can rotate the dials **100** and the clutches **90** will not rotate. The opening-gap **92** of the clutch **90** will maintain the same direction such that after setting a new correct combination code the opening-gap **92** will aligns back to the protrusion **123** of the shackle **120**. In addition, as a result of engagement with the long-leg shackle **122**, the recode indicator **110** has been pushed downward and is being extended out of the lock body **20**/**30** to indicate to the user that the padlock **10** is in the reset mode. The user can rotate the dials **100** to the desired new correct combination code, because the extended fin **91** of each of the clutches **90** have engaged with the fin-slot **20***a*/**30***d* of the lock body **20**/**30**, so that rotation of the dials **100** to the new correct combination code does not also transfer rotation to the clutches **90** as the dials **100** have been disengaged from the corresponding clutches **90**. After setting the new correct combination code, the user can rotate the shackle protrusion **123** ninety (90) degrees such that the recode/reset-protrusion **124** of the shackle **120** will turn away from the recode channel **20***b*/**30***c* of the lock body **20**/**30**. Then the shackle spring **140** is assembled in the spring-loaded area **111** of the recode indicator **110** will push the recode indicator **110** upward and the set of clutches **90** will move upward in the same direction. The extended fin **91** of the clutches **90** will engage back with the teeth **101** of the corresponding dials **100**. The recode indicator **110** will be pushed upward until the large diameter **112** of the recode indicator **110** contacts the top of the recode indicator hole **28**/**39** of the lock body, and the contact clutch surface **113** will push the bottom most clutch **90** upward and the topmost clutch **90** will also contact the wall **29**/**30***a* of the lock body **20**/**30**. The recode indicator **110** contains a long-leg shackle hole receiving hole **114** to receive the long leg **122** of the shackle **120**. To operate the padlock **10** to the locked mode after resetting the correct combination code, the user can place the short-leg **121** of the shackle **120** into the locking-hole **21**/**31** of the body **20**/**30** and scramble the dials **100** such that the opening-gaps **92** of the clutches **90** are away from the protrusions **123** of the shackle **120**.

[0079] It will thus be seen that the objects set forth above, among those made apparent from the preceding description, are efficiently attained and, since certain changes may be made in the above article without departing from the scope of this invention, it is intended that all matter contained in this disclosure or shown in the accompanying drawings, shall be interpreted, as illustrative and not in a limiting sense. It is to be understood that all of the present figures, and the accompanying narrative discussions of corresponding embodiments, do not purport to be completely rigorous treatments of the invention under consideration. It is to be understood that the above-described arrangements are only illustrative of the application of the principles of the present invention. Numerous modifications and alternative arrangements may be devised by those skilled in the art without departing from the scope of the present invention.

Claims

1. A padlock, comprising: a body configured to house a combination mechanism and a key mechanism, the combination mechanism configured to allow operation of the padlock from a locked mode to an unlocked mode by the combination mechanism, and the key mechanism configured to allow operation of the padlock from the locked mode to an unlocked mode by the key mechanism, a shackle having a short-leg and a long-leg, wherein the long-leg is configured for retention in the body and for rectilinear and rotational movement relative to the body of the padlock, a key mechanism indicator configured to at least partially extend out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism, and a latch configured for operative engagement with the key mechanism indicator to retain the key mechanism indicator at least partially extended out of the body of the padlock until the padlock has been operated from the locked mode to the unlocked mode by the combination mechanism.

2. The padlock according to claim 1, wherein the combination mechanism comprises at least one dial having a plurality of indicia thereon, and at least one clutch configured for operative engagement with the at least one dial such that rotation of each of the at least one dial is transferred to the corresponding at least one clutch, wherein each dial of the at least one dial comprises a plurality of teeth, and each clutch of the at least one clutch comprises an extended fin configured for operative engagement with the plurality of teeth of the corresponding dial from the at least one dial.

3. The padlock according to claim 1, wherein the key mechanism comprises a cylinder configured for receipt of a correct cut key and a cam configured for operative engagement with the cylinder such that rotation of the cylinder is transferred to the cam, wherein the cam comprises a cam slope formed therein, and wherein the padlock further comprises a blocking plate having a pin extending therefrom and configured for operative engagement with the cam slope of the cam such that rotational movement of the cam is translated to rectilinear movement of the blocking plate.

4. The padlock according to claim 3, wherein the cam further comprises a slope-wing extending therefrom, and the key-mechanism indicator comprises a spring-loaded area formed on one end thereof, a bottom slope formed on an end of the key-mechanism indicator opposite the spring-loaded area, and a neck formed in the key-mechanism indicator disposed between the spring-loaded area and the bottom slope, wherein the padlock further comprises a key-mechanism indicator spring configured for operative engagement with the spring-loaded area of the key-mechanism indicator to urge the key-mechanism indicator into the body of the padlock, and wherein the slope-wing is configured for operative engagement with the bottom slope of the key-mechanism indicator such that rotation of the cam causes compression of the key-mechanism indicator spring and extension of at least a portion of the key-mechanism indicator out of the body of the padlock.

5. The padlock according to claim 4, wherein the latch comprises a spring assembled area configured for operative engagement with a latch spring and a tip extending from the spring assembled area, wherein the tip is configured for a retaining engagement with the neck of the key-mechanism indicator so as to maintain extension of at least the portion of the key-mechanism indicator out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism.
6. The padlock according to claim 3, further comprising a locking hole formed in the body and configured to at least partially surround the short-leg of the shackle when the padlock is in the locked mode, wherein the locking hole has an arcuate opening formed therein configured for receipt of at least a portion of the blocking plate when the padlock is in the locked mode, wherein the blocking plate is rectilinearly positionable between an extended position when the padlock is in the locked mode and a retracted position when the padlock is in the unlocked mode by the key mechanism as a result of rotational movement of the cam, and wherein when the blocking plate is in the retracted position the shackle may be rotated about the long-leg of the shackle to remove the short-leg of the shackle from the locking hole and operate the padlock to the unlocked mode by the key mechanism.
7. The padlock according to claim 3, wherein the cylinder comprises a square slot, and the cam further comprises a square block configured to be received within the square slot of the cylinder so that rotational movement of the cylinder is transferred to the cam.
8. The padlock according to claim 6, wherein the blocking plate further comprises a blocking-edge, wherein when the blocking plate is in the extended position the blocking-edge is positioned in the arcuate opening of the locking hole to prevent movement of the short-leg of the shackle through the arcuate opening of the locking hole, and wherein when the blocking plate is in the retracted position the blocking-edge is removed from the arcuate opening of the locking hole.
9. The padlock according to claim 2, wherein the long-leg of the shackle comprises a protrusion extending therefrom for each clutch of the at least one clutches, and wherein each clutch of the at least one clutches further comprises a shackle hole configured to receive the long-leg of the shackle through the clutch, a top surface positioned within the shackle hole, a bottom surface positioned opposite the top surface and an opening gap extending from the top surface to the bottom surface and forming a passageway therebetween for passage of the protrusion of the long-leg of the shackle corresponding to the clutch at least partially through the clutch.
10. The padlock according to claim 9, further comprising a locking hole formed in the body and configured to at least partially surround the short-leg of the shackle when the padlock is in the locked mode, wherein rectilinear movement of the long-leg of the shackle away from the body of the padlock is configured to remove the short-leg of the shackle from the locking hole when the padlock is operated to the unlocked mode by the combination mechanism, and wherein the padlock is in the unlocked mode by the combination mechanism when a correct combination code has been set on all of the dials of the at least one dial and the opening gap of each clutch are aligned with each corresponding protrusion of the long-leg of the shackle allowing for movement of each protrusion from the corresponding bottom surface to the corresponding top surface of each corresponding clutch.
11. The padlock according to claim 10, wherein the long-leg of the shackle further comprises a recode protrusion extending therefrom, wherein when the padlock is in the unlocked mode by the combination mechanism the recode protrusion is configured for alignment with a notch formed in the body and insertion of the recode protrusion into the notch engages each protrusion of the long-leg of the shackle with the corresponding top surface of each corresponding clutch thereby causing separation of the extended fin of each clutch from the teeth of the corresponding dial, and wherein the extended fin of each clutch is retained in a fin slot within the body.
12. The padlock according to claim 11, further comprising a recode indicator having a shackle receiving hole configured for receipt of the long-leg of the shackle, a spring loaded area configured

for insertion in a shackle spring and a clutch contact surface, wherein insertion of the recode protrusion into the notch and engagement of each protrusion of the long-leg of the shackle with the corresponding top surface of each corresponding clutch causes operative contact between the bottom surface of at least one clutch with the clutch contact surface and depression of the shackle spring, and wherein depression of the shackle spring allows for at least partial extension of the spring loaded area of the recode indicator out of the body of the padlock.

13. The padlock according to claim 10, wherein the long-leg of the shackle further comprises a reset protrusion extending therefrom, wherein the key-mechanism indicator comprises a spring-loaded area formed on one end thereof and a neck formed in the key-mechanism indicator, wherein the padlock further comprises a key-mechanism indicator spring configured for operative engagement with the spring-loaded area of the key-mechanism indicator to urge the key-mechanism indicator into the body of the padlock, wherein the latch comprises a spring assembled area configured for operative engagement with a latch spring, a tip extending from the spring assembled area and a tail slope extending from the latch opposite the tip, wherein the tip is configured for a retaining engagement with the neck of the key-mechanism indicator so as to maintain extension of at least the portion of the key-mechanism indicator out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism, and wherein the reset protrusion is configured for operative engagement with the tail slope of the latch to urge the latch to compress the latch spring and disengage the tip from the neck of the key-mechanism indicator when the padlock is operated to the unlocked mode by the combination mechanism.

14. The padlock according to claim 12, wherein at least partial extension of the spring loaded area of the recode indicator out of the body of the padlock provides a visual indication the padlock has been operated to a reset mode, wherein the in reset mode rotation of each dial is not transferred to the clutch corresponding to each of the at least one dials.

15. The padlock according to claim 11, wherein the body comprises at least one recode channel formed therein integrally connected with the notch, wherein the recode channel is configured for receipt and retention of the recode protrusion of the long-leg of the shackle within the body after the recode protrusion has been inserted through the notch, and wherein when the recode protrusion is retained within the body by the recode channel separation of the extended fin of each clutch from the teeth of the corresponding dial is maintained.

16. The padlock according to claim 11, wherein the long-leg of the shackle further comprises a reset protrusion extending therefrom, wherein the key-mechanism indicator comprises a spring-loaded area formed on one end thereof and a neck formed in the key-mechanism indicator, wherein the padlock further comprises a key-mechanism indicator spring configured for operative engagement with the spring-loaded area of the key-mechanism indicator to urge the key-mechanism indicator into the body of the padlock, wherein the latch comprises a spring assembled area configured for operative engagement with a latch spring, a tip extending from the spring assembled area and a tail slope extending from the latch opposite the tip, wherein the tip is configured for a retaining engagement with the neck of the key-mechanism indicator so as to maintain extension of at least the portion of the key-mechanism indicator out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism, and wherein the reset protrusion is configured for operative engagement with the tail slope of the latch to urge the latch to compress the latch spring and disengage the tip from the neck of the key-mechanism indicator when the padlock is operated to the unlocked mode by the combination mechanism.

17. The padlock according to claim 13, wherein the long-leg of the shackle further comprises a recode protrusion extending therefrom, wherein when the padlock is in the unlocked mode by the combination mechanism the recode protrusion is configured for alignment with a notch formed in the body and insertion of the recode protrusion into the notch engages each protrusion of the long-leg of the shackle with the corresponding top surface of each corresponding clutch thereby causing separation of the extended fin of each clutch from the teeth of the corresponding dial, and wherein the extended fin of each clutch is retained in a fin slot within the body.

18. The padlock according to claim 11, wherein the key-mechanism indicator comprises a spring-loaded area formed on one end thereof and a neck formed in the key-mechanism indicator, wherein the padlock further comprises a key-mechanism indicator spring configured for operative engagement with the spring-loaded area of the key-mechanism indicator to urge the key-mechanism indicator into the body of the padlock, wherein the latch comprises a spring assembled area configured for operative engagement with a latch spring, a tip extending from the spring assembled area and a tail slope extending from the latch opposite the tip, wherein the tip is configured for a retaining engagement with the neck of the key-mechanism indicator so as to maintain extension of at least the portion of the key-mechanism indicator out of the body of the padlock when the padlock is operated to the unlocked mode by the key mechanism, and wherein the recode protrusion is configured for operative engagement with the tail slope of the latch to urge the latch to compress the latch spring and disengage the tip from the neck of the key-mechanism indicator when the padlock is operated to the unlocked mode by the combination mechanism.

19. The padlock according to claim 13, wherein when the padlock is in the unlocked mode by the combination mechanism the reset protrusion is configured for alignment with a notch formed in the body and insertion of the reset protrusion into the notch engages each protrusion of the long-leg of the shackle with the corresponding top surface of each corresponding clutch thereby causing separation of the extended fin of each clutch from the teeth of the corresponding dial, and wherein the extended fin of each clutch is retained in a fin slot within the body.

20. The padlock according to claim 1, further comprising a locking hole formed in the body and configured to at least partially surround the short-leg of the shackle when the padlock is in the locked mode, wherein rectilinear movement of the long-leg of the shackle away from the body of the padlock is configured to remove the short-leg of the shackle from the locking hole when the padlock is operated to the unlocked mode by the combination mechanism, and wherein rotational movement of the shackle about the long-leg of the shackle is configured to remove the short-leg of the shackle from the locking hole when the padlock is operated to the unlocked mode by the key mechanism.